

Spring-Energy Clustering Is Kernel Clustering:

Known Equivalences Behind a Null Screening

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Abstract

A research line on “mechanical analogies for data clustering” screened clustering objectives built from springs, tension and lifting against matched kernel controls and found no mechanical component that survived the comparison. No manuscript or code of that screening was available to us. We reconstruct the family from the line’s public summary as *pairwise* spring energies (an arbitrary pair potential $\phi(d)$, optionally after a lift, with the within-cluster energy normalised per particle or summed) and read “matched kernel control” as kernel k -means on the same matrix. For that family the null result follows from known results, which we collect and restate for springs: the per-particle spring energy is the pairwise-clustering cost of Hofmann and Buhmann with dissimilarities $\phi(d_{ij})$, hence, up to a constant and a factor $\frac{1}{2}$, a kernel k -means objective that a constant shift makes positive semidefinite without changing the minimisers (Roth et al.; Dhillon, Guan and Kulis); zero-rest-length Hooke springs give k -means; for potentials of negative type, such as the string under constant tension $\phi(d) = d$, the energy is the within-cluster dispersion of energy statistics, the kernel is the dataset-independent distance-induced kernel of Sejdinovic et al., and clustering is kernel k -means in its feature space (França, Rizzo and Vogelstein); the total energy and its tension dual are min-sum k -clustering and max- k -cut; and zero-temperature relaxation is Hartigan’s method in kernel space (kernel k -groups), whose fixed points are Voronoi-stable, i.e. Lloyd fixed points with ties broken in favour of the current cluster. Our own contribution is small: the synthesis and its reading for the research line; the explicit mechanism by which a diagonal shift freezes Lloyd’s rule while leaving the energy and the single-move dynamics unchanged; a proof that the rest-length spring is not of negative type and has no positive definite kernel in the invariance class; and exact rational certificates, on two explicit configurations, that a triangle-area three-body term and a cluster-mean rest length are not pairwise objectives and that a per-spring average is not pairwise under the per-particle or total normalisation. The certificates concern objective values, not minimisers. Computationally, on five datasets and six potentials the spring energies and the kernel objectives coincide to 1.1×10^{-11} on 6090 partitions, a brute-force spring implementation and a kernel implementation of the same relaxation follow identical trajectories in 90/90 runs, and all 600 relaxation fixed points are kernel k -means fixed points. Dynamics on positions is left as an open problem. No claim is made about clustering performance.

1 Introduction

Physical metaphors recur in clustering: algorithms have been built on gravity [32], on spin models at finite temperature [3], on deterministic annealing of a free energy [21, 22], on quantum potentials [16], and on elastic nets [9, 33]. The research line to which this note belongs (LMI001, “Mechanical analogies for data clustering”) studied formulations of a more elementary kind: points connected by springs, with tension and lifting terms, clustered by minimising an elastic energy. Its stated objective was to determine whether the mechanical component produces an effect distinguishable from matched kernel controls, and its screening found none. *The definitions*

of Section 2 are a reconstruction from the line’s public summary; no manuscript or code of the original screening was available to us, and the conclusions below apply to objectives of that pairwise form. We likewise read “matched kernel control” as kernel k -means on the same affinity matrix with the same optimiser; this too is a reconstruction. The purpose of the note is to explain the null result rather than to repeat it, and the explanation is not new: for objectives of the pairwise form it is already contained in three literatures that the line’s summary does not mention. Pairwise data clustering [15] minimises exactly the per-particle spring energy for a dissimilarity matrix (by deterministic annealing, a finite-temperature treatment of the same energy); its reduction to k -means after a constant shift is the constant-shift embedding of Roth et al. [23]; the same objective is the ratio association that weighted kernel k -means optimises [7, 8]. For potentials of negative type, energy statistics supplies a dataset-independent kernel [28], and clustering by within-cluster energy is kernel k -means, optimised preferably by Hartigan’s method [10]. A control that matches the kernel therefore matches the objective exactly, and no experiment on objective values can separate them (experiments on optimisers can, Section 5).

Contribution. The note collects these results in the language of springs, with explicit constants, and draws the consequence for the research line (Section 3). Its own additions are modest: the explicit mechanism by which a diagonal shift freezes Lloyd’s rule while the energy and the single-move relaxation are unchanged (Proposition 3.6), quantified in E3; a short proof that the rest-length spring is not of negative type (Proposition 3.3); exact rational certificates that three specific functionals (a triangle-area three-body term, a cluster-mean rest length and a per-spring average) are not pairwise objectives under the per-particle or total normalisation (Section 4; these concern objective values, not minimisers); and a computational confirmation of the identities with predefined criteria (Section 5). The new mathematical content is small; the value of the note, if any, is in making the null screening a corollary of known results rather than an empirical finding.

Scope. The note is about objectives and their minimisers, not about clustering quality. We do not claim that any potential clusters real data well, and the adjusted Rand indices reported are context, not results. The datasets are standard synthetic ones and two small scikit-learn datasets [20]. Every number in the text is generated by the scripts described in Appendix A, with fixed seeds. Section 6 lists all claims with their status.

2 The pairwise spring family

Throughout, $X = (x_1, \dots, x_n)$ is a configuration in $\mathbb{R}^d$, $k \geq 2$ is fixed, and a *partition* $C = (C_1, \dots, C_k)$ is a partition of $[n]$ into k non-empty clusters, with sizes $n_c = |C_c|$ and indicator vectors $\mathbf{1}_c \in \{0, 1\}^n$. We write $D_{ij} = \|x_i - x_j\|$. “PSD” always refers to a matrix (positive semidefinite); a *kernel* on a set is called *positive definite* (PD) when all its Gram matrices are PSD, the convention of Berg et al. [2].

Definition 2.1 (Spring network, elastic energy, variants). A *spring network* is a triple (ϕ, L, w) of a *pair potential* $\phi: [0, \infty) \rightarrow \mathbb{R}$, a *lift* $L: \mathbb{R}^d \rightarrow \mathbb{R}^D$ (the identity by default), and a *normalisation* $w \in \{w_{\text{sp}}, w_{\text{tot}}\}$ with $w_{\text{sp}}(m) = 1/m$ (energy per particle of the cluster) and $w_{\text{tot}}(m) = 1$ (total energy). Its *within-cluster elastic energy* is

$$E_{\phi, L, w}(C) = \sum_{c=1}^k w(n_c) \sum_{\substack{i < j \\ i, j \in C_c}} \phi(\|Lx_i - Lx_j\|). \quad (1)$$

We write $A_{ij} = \phi(\|Lx_i - Lx_j\|)$ for the *spring matrix*, E_{sp} and E_{tot} for the two normalisations, and $\text{SSE}(C) = \sum_c \sum_{i \in C_c} \|x_i - \mu_c\|^2$, μ_c the centroid, for the k -means objective [18, 19]. The *released*

tension of C is the energy of the springs that cross clusters, $T(C) = \sum_{c < c'} \sum_{i \in C_c, j \in C_{c'}} A_{ij}$; a tension formulation maximises T . An *anchored* network attaches every particle to a hub $y_c \in \mathbb{R}^D$ of its cluster, with energy $E^{\text{anch}}(C, y) = \sum_c \sum_{i \in C_c} \phi(\|Lx_i - y_c\|)$, minimised over the hubs as well as over C .

The potentials used below are: the Hooke spring with zero rest length, $\phi(d) = d^2$ (the spring constant is absorbed); the Hooke spring with rest length $r > 0$, $\phi(d) = (d - r)^2$; the string under constant tension, $\phi(d) = d$, whose per-particle energy is the within-cluster dispersion of energy statistics [10]; the saturating (breakable) spring, $\phi(d) = 1 - e^{-d^2/2\ell^2}$; the soft spring, $\phi(d) = \log(1 + d^2/\ell^2)$; and the Hooke spring on the paraboloid lift $L(x) = (x, \|x\|^2/s)$.

Definition 2.2 (Kernel k -means objective, trace form). For a symmetric matrix $K \in \mathbb{R}^{n \times n}$ and a partition C ,

$$J_K(C) = \text{tr } K - \sum_{c=1}^k \frac{\mathbf{1}_c^\top K \mathbf{1}_c}{n_c}. \quad (2)$$

When K is PSD with a Gram factorisation $K_{ij} = \langle \Phi_i, \Phi_j \rangle$, $J_K(C) = \sum_c \sum_{i \in C_c} \|\Phi_i - m_c\|^2$ with m_c the centroid of the Φ_i , $i \in C_c$: the k -means objective in feature space [6, 12, 27, 34]. For general symmetric K we take (2) as the definition.

Definition 2.3 (Negative type, Bernstein functions). A symmetric kernel ψ on a set S is *conditionally negative definite* (CND) if $\sum_{i,j} c_i c_j \psi(s_i, s_j) \leq 0$ for all finite families $s_i \in S$ and reals c_i with $\sum_i c_i = 0$. A function $\phi: [0, \infty) \rightarrow \mathbb{R}$ is *of negative type* if $(x, y) \mapsto \phi(\|x - y\|)$ is CND on $\mathbb{R}^d$ for every d . A function $\psi: [0, \infty) \rightarrow [0, \infty)$ is a *Bernstein function* if it is continuous, C^∞ on $(0, \infty)$, and $(-1)^{m-1} \psi^{(m)} \geq 0$ for all $m \geq 1$; equivalently (Lévy–Khintchine representation, 25, Thm. 3.2) $\psi(s) = a + bs + \int_{(0, \infty)} (1 - e^{-ts}) \mu(dt)$ with $a, b \geq 0$ and $\int \min(1, t) \mu(dt) < \infty$.

3 Results

3.1 Invariances of the trace form

Lemma 3.1 (What J does not see). *Let K be symmetric, C a partition into k non-empty clusters, $\sigma, a \in \mathbb{R}$ and $f \in \mathbb{R}^n$. Then*

$$J_{K+\sigma I}(C) = J_K(C) + \sigma(n - k), \quad J_{K+a\mathbf{1}\mathbf{1}^\top}(C) = J_K(C), \quad J_{K+f\mathbf{1}^\top + \mathbf{1}f^\top}(C) = J_K(C).$$

Proof. $\text{tr}(\sigma I) = \sigma n$ and $\mathbf{1}_c^\top \sigma I \mathbf{1}_c / n_c = \sigma$ for each of the k clusters. $\text{tr}(a\mathbf{1}\mathbf{1}^\top) = an$ and $\sum_c an_c^2 / n_c = an$. $\text{tr}(f\mathbf{1}^\top + \mathbf{1}f^\top) = 2\sum_i f_i$ and $\mathbf{1}_c^\top (f\mathbf{1}^\top + \mathbf{1}f^\top) \mathbf{1}_c / n_c = 2\sum_{i \in C_c} f_i$, which sums to $2\sum_i f_i$ over c . $\square$

In particular the minimisers of J_K over partitions into k clusters are unchanged by a diagonal shift, by a constant added to every entry, and by a separable term $f_i + f_j$. The lemma is elementary and not new: the diagonal shift is the constant-shift embedding of Roth et al. [23] (a constant added to the off-diagonal dissimilarities) and the device of Dhillon et al. [7, 8] for making an indefinite affinity matrix PSD without changing the optimal partitions; the other two invariances are what connects the spring energy to Schoenberg’s theory below.

3.2 Springs are kernels

Theorem 3.2 (Spring-energy clustering is kernel clustering). *Let (ϕ, L, w) be a spring network with spring matrix A and let $K = -A$ (diagonal included, $K_{ii} = -\phi(0)$). For every partition C into k non-empty clusters:*

(a) $E_{\phi, L, w}(C)$ depends on the data only through A .

- (b) (per-particle normalisation) $E_{\text{sp}}(C) = \frac{1}{2}J_K(C) + \frac{1}{2}(n-k)\phi(0)$. For every $\sigma \geq -\lambda_{\min}(K)$ the matrix $K + \sigma I$ is PSD and $E_{\text{sp}}(C) = \frac{1}{2}J_{K+\sigma I}(C) + \frac{1}{2}(n-k)(\phi(0) - \sigma)$; hence $\arg \min_C E_{\text{sp}} = \arg \min_C J_{K+\sigma I}$, the k -means problem in the feature space of any Gram factorisation of $K + \sigma I$.
- (c) (Hooke) If $\phi(d) = d^2$ and L is the identity, $E_{\text{sp}}(C) = \text{SSE}(C)$ exactly.
- (d) (total normalisation and tension) $E_{\text{tot}}(C) + T(C) = \sum_{i < j} A_{ij}$ is independent of C ; minimising E_{tot} is maximising the released tension, which is the max- k -cut, equivalently min-sum k -clustering, problem with weights A_{ij} [24]. If all clusters have the same size, $E_{\text{tot}} = (n/k)E_{\text{sp}}$.
- (e) (converse) Every kernel k -means objective is a Hooke energy: if K is PSD and $K = \Phi\Phi^\top$, then $J_K(C) = E_{d^2, \Phi, \text{sp}}(C)$, the per-particle Hooke energy of the lifted configuration $(\Phi_1, \dots, \Phi_n)$.
- (f) (universal kernel) If ϕ is of negative type, then

$$\widetilde{K}(x, y) = \phi(\|x\|) + \phi(\|y\|) - \phi(\|x - y\|) - \phi(0)$$

is a PD kernel on $\mathbb{R}^D$ that does not depend on the dataset (given the scales ℓ, r, s that enter ϕ and L), and $E_{\text{sp}}(C) = \frac{1}{2}J_{\widetilde{K}}(C) + \frac{1}{2}(n-k)\phi(0)$ where $\widetilde{K}$ is evaluated on the lifted points. If Φ is a feature map of $\widetilde{K}$ and $\tilde{\Phi} = \Phi/\sqrt{2}$, then $\|\tilde{\Phi}(x) - \tilde{\Phi}(y)\|^2 = \phi(\|x - y\|) - \phi(0)$ and

$$E_{\text{sp}}(C) = E_{d^2, \tilde{\Phi} \circ L, \text{sp}}(C) + \frac{1}{2}(n-k)\phi(0) :$$

up to the constant, the spring energy is the per-particle Hooke energy of the lifted configuration $(\tilde{\Phi}(Lx_i))_i$ (the lift $\tilde{\Phi} \circ L$ takes values in a possibly infinite-dimensional feature space; restricted to the n lifted points it has finite rank, so Definition 2.1 applies with finite D). Without the rescaling, E_{sp} is half the Hooke energy of $(\Phi(Lx_i))_i$ plus the constant.

Status. The theorem collects known results; we state it for springs because the explicit constants are what E1 tests. (a), (d) elementary; (c) classical; (e) the feature-space reading of k -means [12, 34] [classical]. (b) [literature]: with zero diagonal, E_{sp} is the pairwise-clustering cost of Hofmann and Buhmann [15] for the dissimilarities A_{ij} , and its reduction to k -means after a constant shift is the constant-shift embedding of Roth et al. [23]; equivalently, $\sum_c \mathbf{1}_c^\top A \mathbf{1}_c / n_c$ is the ratio association of A , whose maximisation is kernel k -means with $K = \sigma I + A$ [7, 8], and minimising E_{sp} maximises the ratio association of $-A$, i.e. uses $K = \sigma I - A$. Specific to springs is only the constant $\frac{1}{2}(n-k)\phi(0)$. (f) [literature]: $\frac{1}{2}\widetilde{K}$ is the distance-induced kernel of Sejdinovic et al. [28] for the semimetric of negative type $\rho(x, y) = \phi(\|x - y\|) - \phi(0)$ with base point 0 (it is ≥ 0 because the CND test on two points forces $\phi(d) \geq \phi(0)$), $\tilde{\Phi}$ is its feature map, $E_{\text{sp}} - \frac{1}{2}(n-k)\phi(0)$ is the within-cluster energy dispersion of energy statistics for ρ , and its equivalence with kernel k -means is in França et al. [10]; we include the short argument.

Proof. (a) is the definition. (b) Since $\sum_{i < j \in C_c} A_{ij} = \frac{1}{2}(\mathbf{1}_c^\top A \mathbf{1}_c - n_c \phi(0))$,

$$E_{\text{sp}}(C) = \frac{1}{2} \sum_c \frac{\mathbf{1}_c^\top A \mathbf{1}_c}{n_c} - \frac{1}{2}k\phi(0), \quad \text{and} \quad \sum_c \frac{\mathbf{1}_c^\top A \mathbf{1}_c}{n_c} = - \sum_c \frac{\mathbf{1}_c^\top K \mathbf{1}_c}{n_c} = J_K(C) - \text{tr } K,$$

with $\text{tr } K = -n\phi(0)$, so $E_{\text{sp}}(C) = \frac{1}{2}J_K(C) + \frac{1}{2}(n-k)\phi(0)$. The shifted form follows from Lemma 3.1, and $K + \sigma I \succeq 0$ for $\sigma \geq -\lambda_{\min}(K)$. (c) For any finite set P with centroid μ , $\sum_{i, j \in P} \|x_i - x_j\|^2 = 2|P| \sum_{i \in P} \|x_i - \mu\|^2$ (expand $\|(x_i - \mu) - (x_j - \mu)\|^2$; the cross terms vanish). Halving and dividing by $|P|$ gives $\frac{1}{n_c} \sum_{i < j \in C_c} d_{ij}^2 = \sum_{i \in C_c} \|x_i - \mu_c\|^2$; sum over c . (d) Every pair $\{i, j\}$ is either within a cluster or across, so $E_{\text{tot}} + T$ is the total spring energy; the equal-size statement is $w_{\text{tot}} = (n/k)w_{\text{sp}}$. (e) By (b) with $\phi(0) = 0$ applied to the configuration Φ :

$E_{d^2, \Phi, \text{sp}}(C) = \frac{1}{2} J_{-D_{\Phi}^{(2)}}(C)$, where $(D_{\Phi}^{(2)})_{ij} = K_{ii} + K_{jj} - 2K_{ij}$, so $-D_{\Phi}^{(2)} = 2K - f\mathbf{1}^\top - \mathbf{1}f^\top$ with $f_i = K_{ii}$, and Lemma 3.1 gives $\frac{1}{2} J_{2K}(C) = J_K(C)$. (f) Positive definiteness of $\widetilde{K}$ is the lemma relating CND and PD kernels [2, Ch. 3, §2, Lemma 2.1]: for points $x_1, \dots, x_m$ and reals c_i , apply the CND inequality to the family augmented by $x_0 = 0$, $c_0 = -\sum_{i \geq 1} c_i$; substituting c_0 it reads $\sum_{i,j \geq 1} c_i c_j [\phi(\|x_i - x_j\|) - \phi(\|x_i\|) - \phi(\|x_j\|) + \phi(0)] \leq 0$, i.e. $\sum_{i,j} c_i c_j \widetilde{K}(x_i, x_j) \geq 0$. On the dataset, $\widetilde{K} = K + f\mathbf{1}^\top + \mathbf{1}f^\top - \phi(0)\mathbf{1}\mathbf{1}^\top$ with $f_i = \phi(\|Lx_i\|)$, so $J_{\widetilde{K}} = J_K$ by Lemma 3.1 and the identity follows from (b). Next, $\|\Phi(x) - \Phi(y)\|^2 = \widetilde{K}(x, x) + \widetilde{K}(y, y) - 2\widetilde{K}(x, y) = 2\phi(\|x - y\|) - 2\phi(0)$, so $\|\widetilde{\Phi}(x) - \widetilde{\Phi}(y)\|^2 = \phi(\|x - y\|) - \phi(0)$. Finally (e) applied to $\widetilde{K}$ gives $J_{\widetilde{K}}(C) = E_{d^2, \Phi \circ L, \text{sp}}(C)$, and scaling a configuration by $1/\sqrt{2}$ halves every squared distance, so $\frac{1}{2} J_{\widetilde{K}}(C) = E_{d^2, \widetilde{\Phi} \circ L, \text{sp}}(C)$. For Hooke springs $\widetilde{K}(x, y) = 2\langle x, y \rangle$, $\Phi(x) = \sqrt{2}x$ and $\widetilde{\Phi}(x) = x$. $\square$

Proposition 3.3 (Which springs are of negative type). (i) $\phi(d) = d^2$ is of negative type: for $\sum_i c_i = 0$, $\sum_{i,j} c_i c_j \|x_i - x_j\|^2 = -2\|\sum_i c_i x_i\|^2 \leq 0$. [classical]

(ii) If N is a CND kernel with $N \geq 0$ and ψ is a Bernstein function, then $\psi \circ N$ is CND. [literature]

(iii) Consequently $\phi(d) = \psi(d^2)$ is of negative type for every Bernstein ψ ; this covers $d^{2\alpha}$ for $0 < \alpha \leq 1$ (in particular d), $\log(1 + d^2/\ell^2)$, $1 - e^{-d^2/2\ell^2}$ and $d^2/(1 + d^2/\ell^2)$. The rest-length potential $(d - r)^2$ is not of negative type. [proved here (elementary)]

Proof sketch. (i) Expand $\|x_i - x_j\|^2 = \|x_i\|^2 + \|x_j\|^2 - 2\langle x_i, x_j \rangle$; the first two terms vanish against $\sum c_i = 0$. (ii) is proved in Berg et al. [2, Ch. 3, §2] (Schoenberg’s theorem is Theorem 2.2 there; the composition with Bernstein functions follows from it and the representation in Definition 2.3), see also Schilling et al. [25]; we do not vouch for the exact numbering of the composition result. The argument is: by Schoenberg’s theorem [26] N is CND iff e^{-tN} is PD for all $t > 0$, so $1 - e^{-tN}$ is CND (a constant kernel is CND-neutral and the negative of a PD kernel is CND); write $\psi(N) = a + bN + \int (1 - e^{-tN})\mu(dt)$ and note that non-negative combinations and pointwise limits of CND kernels are CND. (iii) The listed ψ are Bernstein: s^α ($0 < \alpha \leq 1$), $\log(1 + s/\ell^2)$, $1 - e^{-s/2\ell^2}$, $s/(1 + s/\ell^2)$ all have completely monotone derivatives. If ϕ is of negative type, the CND inequality for two points with $c = (1, -1)$ gives $2\phi(0) - 2\phi(d) \leq 0$, i.e. $\phi(d) \geq \phi(0)$ for all d ; for $(d - r)^2$, $\phi(r) = 0 < r^2 = \phi(0)$. *Rest length.* Theorem 3.2(b) still applies to $(d - r)^2$: its spring energy is the kernel k -means objective of $-(D - r)^{(2)}$ after a diagonal shift, and its minimisers are feature-space k -means minimisers for that dataset. What fails is a dataset-independent PD kernel. $\widetilde{K}$ is not PD, since $\widetilde{K}(x, x) = 2(\phi(\|x\|) - \phi(0)) = -2r^2 < 0$ when $\|x\| = r$. Nor is any kernel $\widetilde{K}(x, y) + f(x) + f(y) + a + \sigma[x = y]$ with f, a, σ fixed: for the collinear points $0, su, 2su, 3su$ (u a unit vector, $s > 0$) and $c = (1, -1, -1, 1)$, the terms in f and a vanish against $\sum_i c_i = 0$ and the quadratic form of the Gram matrix is $\sum_{i,j} c_i c_j (-\phi(d_{ij})) + 4\sigma = -8rs + 4\sigma < 0$ once $s > \sigma/2r$. Whether Lemma 3.1 lists all the transformations that leave J unchanged up to a constant on every dataset, and hence whether some other dataset-independent PD representative exists, we leave open. Accordingly, E1 finds $\widetilde{K}$ indefinite for this potential on every dataset (Table 1). $\square$

Remark 3.4 (Lifting and anchoring). (a) $E_{\phi, L, w}(X) = E_{\phi, \text{id}, w}(LX)$, so a lift replaces A by $\phi(\|Lx_i - Lx_j\|)$ and nothing else; for Hooke springs a lift is exactly the choice of the kernel $\langle Lx, Ly \rangle$, and by Theorem 3.2(e) every PD kernel arises this way [trivial]. (b) For Hooke springs, $\min_y E^{\text{anch}}(C, y) = E_{\text{sp}}(C) = \text{SSE}(C)$, attained at the centroids (Theorem 3.2(c) and the fact that the centroid minimises the sum of squared distances), and the alternating equilibration “hubs move to the equilibrium of their springs; particles re-attach to the hub of least energy” is Lloyd’s algorithm [18] by definition. For a general ϕ , $\min_y \sum_{i \in C_c} \phi(\|x_i - y\|)$ is a Weber-type location problem; with $\phi(d) = d$ the anchored objective is the k -median objective (geometric

medians as hubs) [classical]. We make no claim about whether these anchored objectives are of the form (1).

3.3 Relaxation of the spring system

The mechanical way to minimise (1) is to let the system relax: one particle at a time jumps to the cluster in which its springs store less energy, or, at finite temperature, jumps with Metropolis probability. On assignments this is a known algorithm.

Proposition 3.5 (Zero-temperature relaxation is Hartigan’s method). *Let K' be any PSD matrix with $J_{K'} = J_K + \text{const}$ on partitions (for instance $K + \sigma I$ or, for ϕ of negative type, $\widetilde{K}$), and write $\delta_{K'}(i, c) = K'_{ii} - \frac{2}{n_c}(K'\mathbf{1}_c)_i + \frac{1}{n_c^2}\mathbf{1}_c^\top K'\mathbf{1}_c$ for the squared feature distance from Φ_i to the centroid of cluster c . Moving particle i from its cluster a ($n_a \geq 2$) to cluster c changes E_{sp} by*

$$\Delta E = \frac{1}{2} \left[\frac{n_c}{n_c+1} \delta_{K'}(i, c) - \frac{n_a}{n_a-1} \delta_{K'}(i, a) \right], \quad (3)$$

and the right-hand side does not depend on the representative K' . Consequently:

- (i) the greedy single-particle relaxation of the spring energy is Hartigan’s method [13, 14] applied to the kernel k -means objective;
- (ii) every partition that is stable under single-particle moves is Voronoi-stable in the feature space of every PSD representative K' , i.e. each Φ_i is assigned to a nearest centroid, so it is a fixed point of Lloyd’s algorithm for K' with ties broken in favour of the current cluster (on the relation between such fixed points and local optimality see 29);
- (iii) the converse fails: $X = \{0, 2, 3.4, 3.6\} \subset \mathbb{R}$, $k = 2$, $C = \{\{0, 2\}, \{3.4, 3.6\}\}$ is Voronoi-stable with SSE = 2.02, while moving the point 2 gives SSE = 1.52.

Proof. Write $S_{ic} = (K'\mathbf{1}_c)_i$ and $Q_c = \mathbf{1}_c^\top K'\mathbf{1}_c$. Removing i from a changes Q_a/n_a to $(Q_a - 2S_{ia} + K'_{ii})/(n_a - 1)$, adding it to c changes Q_c/n_c to $(Q_c + 2S_{ic} + K'_{ii})/(n_c + 1)$, and $\Delta J = -\Delta \sum_c Q_c/n_c$. Expanding both differences and comparing with $\delta_{K'}$ gives $\Delta J = \frac{n_c}{n_c+1} \delta_{K'}(i, c) - \frac{n_a}{n_a-1} \delta_{K'}(i, a)$, an algebraic identity valid for any symmetric K' ; $\Delta E = \frac{1}{2} \Delta J$ by Theorem 3.2(b). Independence of the representative: if $J_{K'} = J_K + \text{const}$ on partitions then $\Delta J_{K'} = \Delta J_K$ for every move, and the identity holds for both matrices, so the right-hand side of (3) is the same for every such K' . Explicitly, under $K' \mapsto K' + \sigma I$, $\delta(i, c)$ increases by $\sigma(1 + 1/n_c)$ for $i \notin C_c$ and by $\sigma(1 - 1/n_a)$ for $i \in C_a$, and both terms of (3) increase by σ ; under the other two invariances of Lemma 3.1, δ itself is unchanged. (i) is the definition of Hartigan’s move. (ii) If no move decreases E then $\frac{n_c}{n_c+1} \delta(i, c) \geq \frac{n_a}{n_a-1} \delta(i, a)$ for all $c \neq a$, and since $\frac{n_c}{n_c+1} < 1 < \frac{n_a}{n_a-1}$ this forces $\delta(i, c) \geq \delta(i, a)$ (not necessarily strictly, whence the tie rule). (iii) Centroids 1 and 3.5; the point 2 is at distance 1 from the first and 1.5 from the second; after the move the clusters $\{0\}$ and $\{2, 3.4, 3.6\}$ (centroid 3) have SSE = $0 + 1 + 0.16 + 0.36 = 1.52$; by (3) with $K' = \widetilde{K} = 2\langle x, y \rangle$, $\Delta E = \frac{1}{2}(\frac{2}{3} \cdot 2 \cdot 1.5^2 - 2 \cdot 2 \cdot 1^2) = -0.5$ (equivalently, the bracket of (3), which is $\Delta J_{K'}$ for any symmetric K' , with $K' = \langle x, y \rangle$ and $J_{\langle x, y \rangle} = \text{SSE}$). $\square$

Status. (i) [literature]: Hartigan’s method in kernel space is the kernel k -groups algorithm of França et al. [10], there derived for energy statistics; the representative-free form (3) is elementary. (ii) is the kernel form of the Euclidean inclusion of Telgarsky and Vattani [30] [proved here]; we could not check whether França et al. [10] state it and claim no priority. (iii) is the standard type of example. The consequence for the research line is that “spring relaxation” cannot converge to anything outside the set of kernel k -means fixed points.

Proposition 3.6 (Lloyd’s rule is representation-dependent). *For $K' = K + \sigma I$, $\delta_{K'}(i, c) = \delta_K(i, c) + \sigma(1 + \frac{1-2\cdot[i \in C_c]}{n_c})$: the shift adds $\sigma(1 + 1/n_c)$ to the squared distance from i to every*

other cluster c and only $\sigma(1 - 1/n_a)$ to the distance to its own cluster a . Hence the Voronoi rule “assign i to $\arg \min_c \delta_{K'}(i, c)$ ” favours staying, and as $\sigma \rightarrow \infty$ every partition becomes Voronoi-stable, while E_{sp} and the single-move dynamics are unchanged. [proved here (two lines); explicit form of an effect noted by 7, 8]

Proof. $(\sigma I \mathbf{1}_c)_i = \sigma[i \in C_c]$ and $\mathbf{1}_c^\top \sigma I \mathbf{1}_c = \sigma n_c$; substitute in δ . $\square$

Dhillon et al. [7, 8] already discuss how the diagonal shift affects the practical performance of kernel k -means; the proposition is the explicit mechanism and E3 quantifies it. So “kernel k -means with Lloyd’s iteration” is not one algorithm for a spring energy but one per PSD representative; the single-move relaxation is the representative-free one. For springs of negative type the natural representative is $\widetilde{K}$ of Theorem 3.2(f), which for Hooke springs is $2\langle x, y \rangle$ and reproduces ordinary Lloyd k -means; the experiments use it, and use the naively shifted $-A + \sigma I$ only as a control (E3).

4 What leaves the family

Theorem 3.2 covers every objective that is a sum over within-cluster *pairs* of a fixed function of the data, weighted by a function of the cluster size in $\{1, 1/n_c\}$. An ingredient can escape only by breaking one of these three features: the sum over pairs (many-body terms), the fixedness of the pair function (parameters that depend on the assignment), or the normalisation. A fourth route replaces the minimisation over partitions by a dynamics on positions. For the first three, specific instances can be certified exactly.

Theorem 4.1 (Exact separation on explicit configurations; computer-assisted). *Let P_1 and P_2 be the configurations of 7 points in $\mathbb{Q}^2$*

$$\begin{aligned} P_1 &= \{(5, 10), (6, 2), (4, 4), (2, 2), (3, 7), (0, 6), (8, 8)\}, \\ P_2 &= \{(0, 2), (7, 4), (8, 7), (5, 0), (2, 8), (2, 3), (4, 10)\}, \end{aligned}$$

drawn once from a dedicated generator (seed 20260932), and let $k = 2$, so that each has 63 partitions. Call a partition function F w -pairwise if $F(C) = \sum_c w(n_c) \sum_{i < j \in C_c} A_{ij} + a$ for some symmetric A and constant a (22 unknowns). Then, with squared distances $s_{ij} = \|x_i - x_j\|^2$, the following holds on both configurations (in parentheses the relative residual $\|b - \Pi b\|/\|b\|$ on P_1 ; P_2 , where $b \in \mathbb{Q}^{63}$ lists the values of F and Π is the orthogonal projection onto the w -pairwise functions):

partition function	w_{sp} -pairwise?	w_{tot} -pairwise?
$\sum_c \frac{1}{n_c} \sum d_{ij}^2$ (Hooke, per particle)	yes (0; 0)	no (0.017; 0.015)
$\sum_c \sum d_{ij}^2$ (Hooke, total)	no (0.179; 0.179)	yes (0; 0)
$\sum_c \frac{1}{n_c} \sum (d_{ij}^2 - \rho)^2$, fixed ρ	yes (0; 0)	no (0.026; 0.030)
$\sum_c \frac{1}{n_c} \sum (d_{ij}^2 - \rho_c)^2$, $\rho_c = \text{cluster mean}$	no (0.143; 0.157)	no (0.109; 0.122)
$\sum_c \frac{1}{n_c} \sum_{i < j < l} 4 \text{area}(ijl)^2$ (three-body)	no (0.331; 0.328)	no (0.039; 0.037)
$\sum_c \binom{n_c}{2}^{-1} \sum d_{ij}^2$ (per spring)	no (0.185; 0.180)	no (0.125; 0.121)

where ρ is the mean of all s_{ij} , ρ_c the mean of s_{ij} over the pairs of cluster c , and $\text{area}(ijl)$ the area of the triangle $x_i x_j x_l$. Each row is one function F , tested against both families.

Proof. For fixed w the w -pairwise functions form a linear subspace of $\mathbb{R}^{63}$ spanned by the columns of an explicit rational design matrix (rank 21 for w_{sp} , 22 for w_{tot} , on both configurations). Membership is decided by exact least squares in rational arithmetic (`fractions.Fraction`): a zero residual exhibits A and a ; a non-zero residual is a certificate that F is not in the span. The script `experiments/identity_check.py` (E4) performs the computation; the relative residuals are reported in the table. $\square$

The two positive controls (Hooke and fixed rest length under their own normalisation, total Hooke under w_{tot}) confirm that the test is not blind, and the yes/no pattern agrees on the two configurations in 12 of the 12 cells (a sanity check: one configuration suffices logically). The table says that the two normalisations are different objectives (each fails to be pairwise under the other’s weight), and that the three functionals tested, all with the per-particle weight $1/n_c$ inside F (rows 4–6), are neither w_{sp} - nor w_{tot} -pairwise; their total-weight versions were not tested. The per-spring average *is* a pairwise sum with the size weight $\binom{n_c}{2}^{-1}$; the certificate only excludes the weights 1 and $1/n_c$. By Cauchy–Binet applied to the rows $(1, x_i^\top)$, $\sum_{i < j < l \in C} 4 \text{area}(ijl)^2 = |C| \det S_C$ with S_C the scatter matrix of C , so the three-body row is $\sum_c \det S_c$, a quartic in the data. The theorem certifies these three functionals, not the classes “many-body terms” or “assignment-dependent rest lengths”, and the class statements would be false: restricted to partitions whose clusters all have at least three points, $\sum_c w(n_c)/(n_c - 2) \sum_{i < j < l \in C_c} (s_{ij} + s_{jl} + s_{il}) = \sum_c w(n_c) \sum_{i < j \in C_c} s_{ij}$.

Values, not minimisers. The certificate concerns objective values up to an additive constant. It does not show that the minimisers differ from those of every pairwise objective: with A free, every partition is the unique minimiser of some pairwise objective ($A_{ij} = -1$ within its blocks, 0 otherwise), and a strictly increasing non-affine transform of a pairwise objective fails the test while keeping its minimisers. Whether these ingredients can produce an *observable* mechanical effect is therefore a separate question that the theorem does not answer.

The dynamic route. Several physics-inspired algorithms do not minimise an energy over partitions but move the *positions*: gravitational clustering [32], mean shift [4, 5, 11] (a gradient ascent on a kernel density estimate, hence a dynamics under a pairwise potential), quantum clustering [16], and the finite-temperature correlations of a Potts model [3]. Their output is a function of the trajectory, and Theorem 3.2 says nothing about it.

Open problem 4.2 (Dynamic mechanical clustering). Let $V: (0, \infty) \rightarrow \mathbb{R}$ be a pair potential, let $z(t)$ solve $\dot{z}_i = -\sum_{j \neq i} V'(\|z_i - z_j\|) \frac{z_i - z_j}{\|z_i - z_j\|}$ with $z(0) = X$, and let $P_{T,\varepsilon}(X)$ be the single-linkage partition of $z(T)$ at threshold ε . Characterise the triples (V, T, ε) for which there exist a fixed function $a: [0, \infty) \rightarrow \mathbb{R}$ and a normalisation w such that, for every X , $P_{T,\varepsilon}(X)$ minimises $\sum_c w(n_c) \sum_{i < j \in C_c} a(\|x_i - x_j\|)$ over partitions into the number of clusters it produces.

Conjecture 4.3. *For the Gaussian attraction $V(d) = -e^{-d^2/2h^2}$ and generic (T, ε) no such pair (a, w) exists: the partition then depends on the sample through the basins of the density modes, which are not determined by the pairwise distances weighted independently of the assignment. [conjectural; not tested here]*

Whatever the answer, the dynamic route is already occupied by mean shift, so a mechanical proposal of this kind has to be compared with mean shift, not with k -means, and the comparison must involve the trajectory, not the energy minimum.

5 Computational confirmation

Data and potentials. Five datasets: blobs, two moons and two circles from scikit-learn’s generators (300 points each in $\mathbb{R}^2$; `make_blobs` with three centres and std 0.9, $k = 3$; `make_moons` with noise 0.06 and `make_circles` with factor 0.45 and noise 0.05, $k = 2$), a random subset of 300 digits from `load_digits` (64 features scaled to $[0, 1]$, $k = 10$) and the standardised iris data (150 points, 4 features, $k = 3$). The six potentials of Section 2, with scales set from the data before any run: ℓ is the median distance to the seventh nearest neighbour, r the tenth percentile of all pairwise distances, s the median of $\|x\|^2$. In all, $5 \times 6 = 30$ configurations. Seeds: 20260930 (E1, E2), 20260931 (E3) and 20260932 (E4).

potential	neg. type (Prop. 3.3)	partitions	max rel. error	min $\lambda_{\min}(\tilde{K})/\lambda_{\max}(\tilde{K})$	PSD
d^2 (Hooke)	yes	1015	3.6×10^{-12}	-2.6×10^{-16}	5/5
d^2 on the lift ($x, x ^2/s$)	yes	1015	2.6×10^{-12}	-3.2×10^{-16}	5/5
$(d-r)^2$ (rest length)	no	1015	1.1×10^{-11}	-5.5×10^0	0/5
d (constant tension)	yes	1015	2.2×10^{-13}	-4.7×10^{-18}	5/5
$1 - e^{-d^2/2\ell^2}$ (saturating)	yes	1015	1.2×10^{-13}	≥ 0	5/5
$\log(1 + d^2/\ell^2)$ (soft)	yes	1015	1.4×10^{-13}	≥ 0	5/5

Table 1: E1. Per potential, over the five datasets: whether Proposition 3.3 labels the potential of negative type (a label, not a computation); number of partitions tested; worst relative discrepancy between the spring energy and the kernel objectives; most negative normalised eigenvalue of the universal kernel $\tilde{K}$ across datasets; number of datasets on which $\tilde{K}$ is numerically PSD.

E1: identical objective values. *Criterion (predefined):* for every tested partition and every potential, $|E_{\text{sp}} - (\frac{1}{2}J_{K'} + \text{const})| \leq 10^{-9} \max(1, |E_{\text{sp}}|)$ with the constant of Theorem 3.2, for the three representatives $K' \in \{-A, -A + \sigma I, \tilde{K}\}$; pass means 100%. The partitions are 200 uniformly random ones per configuration, the outputs of Lloyd and of the single-move relaxation on $\tilde{K}$ (minimally shifted when indefinite), and the ground-truth labels. *Result:* 19285/19285 checks passed, worst relative error 1.1×10^{-11} (the largest errors occur with the shifted representative, where σ is up to 4.4×10^4 and the identity involves cancellation); the Hooke energy equals the k -means SSE to 3.9×10^{-16} . On equal-size partitions $E_{\text{tot}} = (n/k)E_{\text{sp}}$ to 2.2×10^{-16} , while on random partitions the ratio $E_{\text{tot}}/E_{\text{sp}}$ ranges over $[29.7, 197.5]$: the normalisation is a genuine choice. Table 1 also reports the spectrum of $\tilde{K}$: for the 25 configurations with a potential of negative type it is PSD in 25 (numerically, $\lambda_{\min}/\lambda_{\max} \geq -3.2 \times 10^{-16}$), and for the rest-length potential it is indefinite in 5/5.

E2: same optimiser, same initialisation, same trajectory. *Criterion:* a brute-force spring relaxation that recomputes the total energy from scratch for every candidate move (it sees only springs) and the kernel implementation of Hartigan’s method via (3) (it sees only K), started from the same random labels with the same sweep order, must produce identical label vectors after every sweep and final energies equal to 10^{-9} relative, in all runs. Subsets of 120 points, three initialisations per configuration. *Result:* 90/90 identical trajectories (mean 4.9 sweeps, at most 11), worst final-energy discrepancy 1.4×10^{-14} ; the brute-force runs took 56 s in total against 0.6 s for the kernel runs. In addition, hub-anchored Hooke springs equilibrated alternately (Remark 3.4(b)) reproduced scikit-learn’s Lloyd k -means from the same initial centroids in 15/15 runs, with inertia equal to the spring energy to 4.3×10^{-16} .

E3: relaxation versus kernel k -means on the same energy. Five optimisers see the same kernel: *lloyd* (batch Lloyd on $\tilde{K}$, minimally shifted only when indefinite, 5 configurations), *lloyd+relax* (Lloyd, then single-move polish), *relax* (single-particle relaxation from the initial labels), *anneal* (Metropolis on the spring energy, 40 sweeps of geometric cooling from the median $|\Delta E|$ of random moves down by 10^{-3} , then relaxation [17]) and the control *lloyd (naive)* (Lloyd on $-A + \sigma I$, Proposition 3.6). Initialisations are shared: 10 random labelings and 10 kernel k -means++ seedings [1] per configuration; annealing uses the first 5 random ones. E^* is the best energy found by any method in a configuration. *Criteria (predefined):* (C3a) every relaxation and annealing fixed point is Voronoi-stable for the kernel used (Proposition 3.5 predicts 100%); (C3b) the relaxation is called distinguishable from the kernel family only if its best-of-restarts energy differs from that of *lloyd+relax* by more than 10^{-6} relative, either sign being reported since it would be a statement about optimisers of one function; (C3c) the fraction of Lloyd fixed points that are not single-move stable is reported. *Disclosure:* a pilot run used $-A + \sigma I$ as the only Lloyd kernel and found it inert (Proposition 3.6); $\tilde{K}$ then replaced it as

potential	configurations reaching E^* (of 5)				restarts reaching E^* (%)			Lloyd f.p.	best
	relax	lloyd+relax	lloyd	naive	relax	lloyd+relax	lloyd	1-move stable	ARI= 1
d^2 (Hooke)	4/5	5/5	3/5	1/5	54	57	31	46/100	4/5
d^2 on the lift ($x, x ^2/s$)	5/5	5/5	4/5	2/5	80	74	69	73/100	5/5
$(d-r)^2$ (rest length)	5/5	4/5	1/5	1/5	53	49	4	4/100	4/5
d (constant tension)	5/5	5/5	2/5	1/5	42	43	30	43/100	5/5
$1 - e^{-d^2/2\ell^2}$ (saturating)	4/5	2/5	2/5	0/5	4	2	2	12/100	2/5
$\log(1 + d^2/\ell^2)$ (soft)	5/5	4/5	2/5	1/5	37	34	17	26/100	4/5

Table 2: E3 by potential (five datasets each). “Reaching E^* ” means within 10^{-6} relative of the best energy found by any method. “Lloyd f.p. 1-move stable”: Lloyd fixed points that no single move improves (all relaxation fixed points are Lloyd fixed points, Proposition 3.5(ii); 100/100 per potential). “best ARI= 1”: configurations in which the best partitions of relax and lloyd+relax coincide.

the baseline, which favours the kernel side of the comparison, and the naive shift was kept as a control. C3a–C3c and their thresholds were written before the pilot and were not changed; every number reported comes from the final code. *Results* (Table 2, Figure 1; per configuration in `results/tables_relaxation.md`): all 600/600 relaxation and 150/150 annealing fixed points are Voronoi-stable (C3a). Under C3b, 6 of 30 configurations are distinguishable (5 in favour of the relaxation, 1 against; largest relative difference 7.4×10^{-3} ; the split is optimiser noise: the `-fast` run gives 3–3 with the same 6/30); whenever the two best energies agree the best partitions coincide (ARI = 1; 0 exceptions), and overall ARI = 1 in 24/30. The relaxation reaches E^* in 28 configurations, lloyd+relax in 25, Lloyd alone in 14 and the naive shift in 6 (annealing, with 5 restarts only, is not comparable on this count: 20); the median relative excess over restarts (median across configurations) is 8.4×10^{-4} , 7.7×10^{-4} , 3.7×10^{-3} and 5.9×10^{-1} . Only 34% of Lloyd fixed points on the canonical kernel are single-move stable, 6% on the naive shift (C3c); the single-move methods are single-move stable in 600/600, 600/600 and 150/150 runs (measured; expected by construction). The best energies were recomputed spring by spring, worst discrepancy 4.3×10^{-13} . E3 is descriptive (counts over 30 configurations, modest restart budgets); no inferential claim is made.

Reading of E1–E3 for the research line. If the screened formulations were of the pairwise kind of Definition 2.1, the screening could not have found a mechanical effect: the spring energies *are* the kernel objectives (E1), the mechanical implementation of the relaxation *is* a kernel implementation (E2), and every relaxed state *is* a kernel k -means fixed point (E3). Differences between optimisers of the same function are visible (Lloyd alone reaches E^* less often than the single-move methods; the naive shift is worse still), and they are the only differences there are.

6 What can be claimed

Table 3 collects the mathematical claims with their status; it separates what closes the screening by proof from what was only measured. The measured items are those of Section 5 (E1: 19285/19285 identities, $\tilde{K}$ PSD in 25/25 negative-type and indefinite in 5/5 rest-length configurations; E2: 90/90 identical trajectories, 15/15 anchored runs equal to Lloyd; E3: 600/600 and 150/150 Voronoi-stable fixed points, 6/30 configurations with different best energies, naive-shift Lloyd at E^* in 6/30 against 14); “verified” means reproduced by the scripts of Appendix A with the fixed seeds, not proved.

Limitations. The family of Section 2 is a reconstruction from the line’s public summary; if the original screening used many-body terms, assignment-dependent parameters or a dynamics on positions, Section 4 applies and the theorem does not. The theorem is about objectives; it does not say that all optimisers of a spring energy behave alike, and E3 shows that they do not.

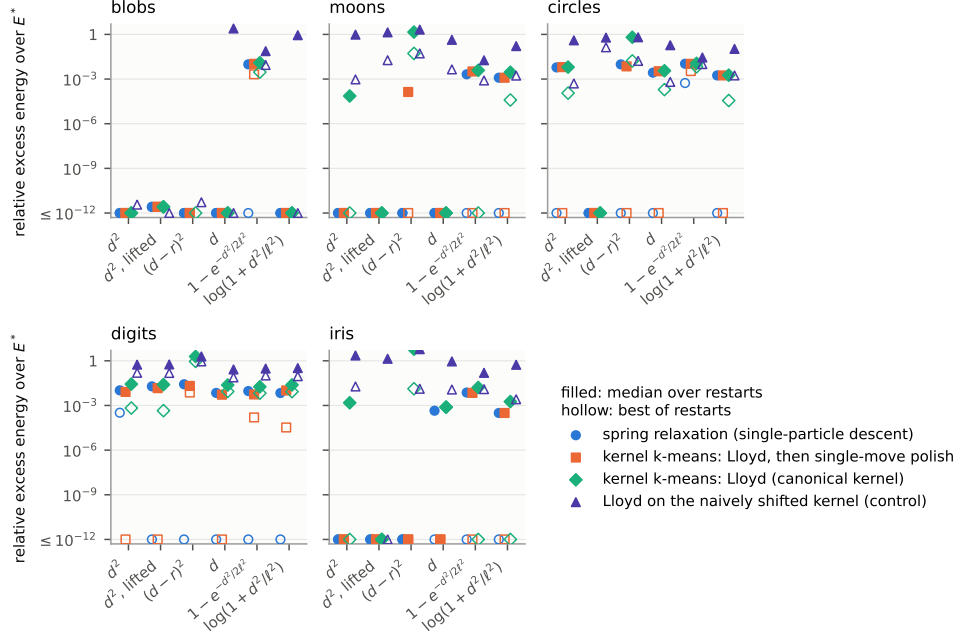


Figure 1: E3. Relative excess energy over E^* for the four optimisers of the same spring energy, per dataset and potential (filled: median over restarts; hollow: best of restarts; values below 10^{-12} are drawn at the floor). The optimisers differ in how often they reach the best local optimum, not in what they optimise; the naively shifted Lloyd is the control of Proposition 3.6.

The normalisation family is $\{1, 1/n_c\}$; other size weights (for instance the per-spring average) leave the exact identity and we only certify that they are not pairwise under the two standard weights. Theorem 4.1 certifies three specific functionals on two configurations, which is what “not in general” statements need, but says nothing about classes of terms or about how far from pairwise those energies are on typical data. The Bernstein composition result is cited, not re-proved in full. E3 uses one implementation of each optimiser with modest restart budgets; its numbers describe optimiser luck on five datasets, not a benchmark. Nothing is claimed about the dynamic route beyond its formulation.

Next steps for the research line. (a) Confirm with the author of the line that the screened formulations were of the pairwise form of Definition 2.1; if not, the emphasis moves to Section 4. (b) Settle Open problem 4.2 at least for the Gaussian flow, either by an explicit configuration where $P_{T,\varepsilon}$ is not a pairwise minimiser (the exact-arithmetic method of Theorem 4.1 applies once $P_{T,\varepsilon}$ is computed), or by a representation. (c) Characterise the size weights w for which $\sum_c w(n_c) \sum_{i < j \in C_c} A_{ij}$ is a weighted kernel k -means objective in the sense of Dhillon et al. [8]; the answer decides whether the normalisation is a mechanical degree of freedom or a graph-cut one. (d) If (a) is confirmed, the line’s summary can be updated from “no mechanical component survived” to “no mechanical component of the pairwise kind can survive”, with the open ingredients named.

A Reproducibility

The directory contains `experiments/mechanical.py` (potentials, energies, kernel forms, optimisers), `experiments/identity_check.py` (E1, E2, E4), `experiments/relaxation.py` (E3), `experiments/make_figures.py` and `experiments/make_numbers.py`, which converts `results/*.json` into the macros and table bodies used here, so that every number in the text is generated; `results/tables_identity.md` and `results/tables_relaxation.md` hold

claim	status
J is invariant under diagonal shift (up to $\sigma(n - k)$), constant and separable terms (Lemma 3.1)	proved here (elementary)
Per-particle spring energy = $\frac{1}{2}$ ratio association $-\frac{1}{2}k\phi(0) = \frac{1}{2}J_{-A} + \text{const}$, with a PSD representative after a shift (Thm. 3.2(b))	literature (ratio association $\leftrightarrow$ kernel k -means, 6, 8), stated here for spring matrices with the explicit constant
Zero-rest-length Hooke energy = k -means SSE (Thm. 3.2(c))	classical
Total energy / released tension = min-sum k -clustering / max- k -cut with weights A (Thm. 3.2(d))	proved here (elementary)
Every kernel k -means objective is a Hooke energy of a lifted configuration (Thm. 3.2(e))	classical (feature-space k -means)
Negative-type springs have a dataset-independent PD kernel and are Hooke springs after a lift and a $1/\sqrt{2}$ rescaling (Thm. 3.2(f))	proved here from Berg et al. [2]
d^2 is of negative type; Bernstein transforms preserve negative type (Prop. 3.3)	classical / literature
Rest-length springs: covered dataset-wise; no universal kernel exhibited ($\tilde{K}$ indefinite on all 5 datasets)	proved / open (verified numerically)
Lifting only changes the kernel; anchored Hooke springs are Lloyd k -means (Remark 3.4)	trivial / classical
Single-particle relaxation is Hartigan’s method; its fixed points are Lloyd fixed points for every PSD representative (Prop. 3.5)	proved here (generalises 30)
Lloyd’s rule depends on the PSD representative; large shifts freeze it (Prop. 3.6)	proved here; explicit form of an effect noted by Dhillon et al. [7, 8]
The cluster-mean rest length, the triangle-area three-body term, the per-spring average and the cross-normalisation tested are not pairwise objectives (Thm. 4.1)	proved here (exact arithmetic, two explicit configurations)
Dynamic (position-relaxation) mechanical clustering is not a pairwise objective (Conj. 4.3)	conjectural
Any statement about clustering quality on real data	not addressed

Table 3: Mathematical claims and their status.

the full tables (E4 on both configurations; E3 per configuration). Both experiment scripts take `-fast`. The reference runs used seeds 20260930 (E1, E2; E4 uses its own generator with seed 20260932, so its configurations do not depend on `-fast`) and 20260931 (E3), Python 3.11.15, NumPy 2.4.6, SciPy 1.17.1 [31] and scikit-learn 1.9.1 [20]; E1, E2 and E4 took 159 s and E3 128 s on shared cores. *Energies and tests.* E_{sp} and E_{tot} are computed from the definition (1) by summing ϕ over the within-cluster pairs of the distance matrix; J_K from (2). Partitions always have k non-empty clusters; a Lloyd step that would empty a cluster refills it with the worst-fitted point. Relaxation sweeps in a fresh random order; a particle moves to the argmin cluster only if the energy decreases by more than 10^{-10} times the initial energy scale, identically in both implementations of E2; a particle alone in its cluster does not move. Annealing proposes a uniformly random other cluster and accepts with probability $\min(1, e^{-\Delta E/T})$. Voronoi stability: own δ within 10^{-9} (relative to the largest $|\delta|$) of the minimum over clusters; single-move stability: no $\Delta E < -10^{-9}$ times the energy scale. *E4.* Integer coordinates drawn uniformly in $\{0, \dots, 10\}^2$ (repeated until distinct) from `default_rng(20260932)`; the design matrix has one column per pair and one for the constant; the normal equations are solved by exact Gaussian elimination over $\mathbb{Q}$.

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